

From Drosophila Male Connectome to Artificial Neural Architectures

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Deep Research

EXECUTIVE SUMMARY

The MaleCNS is the fully proofread central nervous system of an adult *Drosophila melanogaster* male, reconstructed at synaptic resolution using seven eFIB-SEM systems over thirteen months to yield a 160-teravoxel volume. Rather than a functional specification, it is a wiring diagram recording which neurons connect and how many synapses span each pair. The report surveys connectome-inspired machine learning, which uses this connection pattern as raw artificial network wiring, as a structural prior that regularizes training, and as the substrate for graph and recurrent architectures, all while cautioning against biologically unwarranted claims.

The report proposes three increasingly faithful model classes built from the same reconstruction. A coarse cell-type graph model captures neuron-type connectivity; a mesoscopic circuit model preserves the major anatomical pathways and recurrence at the scale of neuron types; and a neuron-level sparse recurrent model restores individual cells with estimated synaptic signs and sensory-motor interfaces. Success in each tier requires specifying data needs, computational cost, dynamical assumptions, training objective, baselines, evaluation tasks, and ablations. Recurrence carries a steep sequential cost, and evaluation must distinguish what success demonstrates from what it cannot.

Two complementary control strategies let agents inherit real circuits: neural-circuit policies treat the connectome's layered wiring as a fixed structural prior, while biologically constrained reinforcement learning grounds the learning rule in mechanisms a brain could plausibly implement. Graph neural networks operate directly on the graph-structured connectivity. Regardless of approach, every model inherits the gap between wiring and function. The report enumerates missing variables—synaptic physiology, neuromodulators, gap junctions, temporal dynamics, developmental history, embodiment, and learning rules—and warns that claims exceeding the MaleCNS's explanatory scope are biologically unwarranted.

MaleCNS connectome anatomy and organization

The MaleCNS connectome is the fully proofread central nervous system of an adult *Drosophila melanogaster* male, reconstructed at synaptic resolution to support end-to-end sensory-to-motor circuit analysis [1]. It was assembled with seven enhanced focused ion beam scanning electron microscopy (eFIB-SEM) systems over

thirteen months, producing a 160 teravoxel image volume acquired at $8 \times 8 \times 8$ nm isotropic resolution [1]. The effort extended the previously published right optic lobe connectome to the complete pair of optic lobes along with the entire central brain and ventral nerve cord [1]. A central organizing claim is that sexual dimorphism exerts brain-wide influence across multiple sensory and behavioral circuit modules rather than residing in a single isolated region [1].

Cell-type taxonomy and cross-dataset annotation

Cell-type annotation formed the backbone of the male CNS resource, with 97.9% of neurons across the CNS matching a cell type in the FAFB/FlyWire, hemibrain, and/or male adult nerve cord (MANC) datasets [1]. The remaining 2.1% were largely unmatched because of the limitations of previous partial datasets rather than genuine novelty [1]. Because nomenclature was inconsistent across references, annotations were recorded in parallel as mancType, flywireType, and hemibrainType, so that the male CNS now serves as a connectomic Rosetta Stone for cross-dataset comparison [1]. Quality control rested on explicit error estimates: expert curation put the error rate below 1% for neurons in each hemisphere, and if errors occur independently in the two hemispheres the probability of missing an entire cell type falls below 0.01%, corresponding to roughly three pairs of neurons in the central brain [1]. That low false-positive rate is set against more than 1,000 male-specific neurons, so these cells reflect real biology rather than missing wiring [1]. The authors stress that expert curation remained essential to separate biological signal from technical noise, which tends to be correlated across both hemispheres of a single brain [1].

Sex classification framework for neurons in the male versus female (FAFB/FlyWire) connectomes.

CELL CATEGORY	DEFINING PROPERTY	SUPPORTING OBSERVATION
Isomorphic	Present in both sexes with matching morphology and connectivity [1].	AVLP016 shown as the closest isomorphic reference for a male example [1].
Dimorphic	Sex differences in morphology or connectivity, e.g. ventrolateral axons present only in the male [1].	Synapses are spatially clustered along feedforward and feedback sensorimotor wiring [1].
Sex-specific	Neurons occurring in only one sex, with more than 1,000 male-specific cells identified [1].	Emerged as the dominant class of sex differences in the resource [1].

Synaptic topology, modularity, and sensorimotor pathways

The male CNS joins a sex-shared sensorimotor framework with a sex-specific overlay that concentrates in higher-order, integrating centers [1]. The analysis spans visual, auditory, olfactory, and taste pathways, all revealing shared organizational principles across sexes [1]. A hierarchy emerges in which sex differences primarily modify integrative and decision-making areas while sensory detection and the highly tuned motor

interface remain comparatively constant [1]. This organization is distinct from *C. elegans*, where male-specific neurons are scarce in the nerve ring [1]. Dimorphic cell types carry spatially clustered synapses that sit within feedforward and feedback connections of particular sensorimotor pathways, such as vision or ascending taste neurons [1]. Downstream of sex-shared VPNs, wiring motifs include sign inversion contributed by male-specific neurons [1]. The structural resource is expressed as a chemical-synapse connectome with excitatory and inhibitory sign estimated from synapse data; electrostatic gap-junction couplings are likewise present in such reconstructions but are not separately enumerated here.

In the olfactory system, a bidirectional circuit switch reroutes pheromone signals between male and female brains [1]. A separate sex-specific switch shifts the balance between visual and olfactory inputs that underlie adaptive sex differences in behavior [1]. Inter-male aggression circuits are among the socially relevant loops the resource exposes [1]. Circuit modules linking internal state to social behavior provide the interpretive backdrop for these sex-dimorphic loops [1].

The visual pathway was reconstructed on a new eye map representing the visual field of the compound eyes, extending the Zhao et al. framework to accommodate medulla column regions of interest in the male brain [1]. Regional cell counting follows a greater-than-2% synapse convention, and many neurons contribute to multiple regions [2].

Modularity of such connectomes is conventionally framed as network science applied to connection adjacency matrices [3,4]. Hierarchical community structure—nested communities linked to cellular annotation—has been documented in the larval *Drosophila* connectome and can be probed with hierarchical block-model selection [1]. Thermosensory and hygrosensory processing extends across first-, second-, and third-order neurons, illustrating multi-tier integration within the fly brain [1].

Within the central complex, the ellipsoid body and pedunculus (EB and PB) were traced with steadily increasing fractions of synapses having identified partners [5]. A systematic annotation of the male nerve cord assigned a sensory modality to more than 5,000 sensory neurons, clustered them by connectivity, and identified serially homologous cell types [6].

Coarse cell-type graph model design

Data requirements and graph construction

Building this graph starts from the MaleCNS connectivity matrix, whose per-neuron skeletons, synapse coordinates, and synapse counts come through neuPrint and FlyWire materialization version 783 [1]. The supporting Python tooling, including fafbseg-py for segmentation and skeletons, is bundled with spatial-transform packages for the various template spaces [1]. Cell types come from morphological co-clustering with NBLAST, where skeletons are pruned to remove twigs shorter than 5 μm , combined with connectivity

clustering, so the coarse model reuses the taxonomy the connectome paper defines rather than inventing one [1]. Each segment is then assigned a neurotransmitter prediction with a confidence score derived from the synapse-label confusion matrix, and that prediction supplies the excitatory, inhibitory, and disynaptic signs attached to edges [1]. The aggregate edge weight between cell types a and b is the summed synapse count over every neuron pair spanning the two types, and that scalar is split between the left and right hemispheres by the source neuron's soma (somaSide); unpaired medial neurons are resolved by target side and neutral connections are divided evenly [1]. Anatomical parcellation of the source neuropils is transferred from the JRC2018M template through the synapse point cloud, and hemispheric homology is enforced by aligning neurons across the midline [1]. Because the graph inherits every error in the source segmentation, the MaleCNS proofreading, in which each neuron was inspected for false merges, repaired as needed, and each soma fiber traced back to its neuron, bounds how trustworthy the aggregate weights are [1]. Reconstruction is uneven, so nodes are weighted or filtered by the neuron-centric downstream capture metric, the fraction of a neuron's synaptic outputs that land on traced partners, which is bimodal across the central-brain compartments [1]. Where groups coarser than the supplied cell-type taxonomy are wanted, a nested stochastic block model (SBM) can define the blocks, placing edges between groups according to block-specific connection probabilities and selecting the best partition by the minimum description length principle [1]. Hierarchical community detection is the tool used to quantify the contributions of isomorphic and sex-specific cell types to network structure, and its model-selection discipline follows the hierarchical block-structure literature [1]. The same low-dimensional UMAP embeddings or Ward's-method linkages used to visualize clustering also sanity-check that the coarse blocks are coherent [1].

$$W_{ab}^c = \sum_{i \in a} \sum_{j \in b} W_{ij} \quad J_{ab} = \frac{W_{ab}^c}{n_a n_b}$$

Here W_{ij} is the synapse count between neurons i and j, n_a is the size of type a, and the rescaled coupling J_{ab} is a mean-field average per neuron pair. The coarse graph therefore reduces N neurons and their sparse interconnections to a dense C x C coupling matrix over cell types. Because C is far smaller than N, the adjacency is compact and the expensive neuron-level sparsity is replaced by a small dense matrix.

Dynamical assumptions and computational cost

The dynamical assumption is mean-field: a cell type is collapsed to a small state, typically a single rate variable, so type-to-type coupling is the aggregate weight rescaled by type population and dynamics reduce to C coupled equations rather than N. Edges are static scalar couplings, and the model deliberately ignores within-type heterogeneity, spike timing, and temporal synaptic effects; those are exactly the variables left to the finer model classes and the missing-variables discussion. Construction is one $O(N)$ aggregation pass over the sparse neuron adjacency, after which memory and simulation are dominated by the $O(C^2)$ coarse matrix. The one-time SBM or community-detection preprocessing, run with the graph-tool library for network analysis and inference, adds

a polynomial pass that never recurs during simulation [1]. Consequently a coarse graph of a few hundred cell types trains and simulates orders of magnitude faster than a neuron-level model while preserving only aggregate routing and motif statistics.

Training objective and baselines

The coarse graph is principally a structural prior, so training either fits a generative model to the aggregate adjacency or conditions a learnable dynamics model on it. As a generative fit the objective is the SBM likelihood maximized under the minimum description length criterion [1]. When the coarse graph constrains a learnable dynamics model, the objective couples a sensorimotor or activity-prediction loss with a structural penalty keeping the learned coupling near the connectome-derived weights, reflecting the finding that task optimization matters relative to random single-cell and synapse parameters [7]. Where the coarse adjacency is treated as discrete categorical data, a discrete diffusion model over connection-weight tokens is a viable objective alternative to autoregressive loss [8]. Baselines must include a random connectome ensemble with independently drawn single-cell and synapse parameters as a structural null [7]. The same assembly should be run at full neuron resolution as the fine-grained comparison, and the female hemibrain connectome should serve as a biological baseline, since its connectivity is reachable through the same neuPrint and FlyWire resources [1]. Because network models routinely draw direct associations between abstract model quantities and anatomically defined neurons, a proper null and a neuron-level competitor are essential before any coarse-graph advantage is credited to the cell-type collapse itself [9].

Evaluation tasks and ablations

Evaluation centers on sensorimotor information flow through the type-level graph, recovery of the isomorphic versus dimorphic community structure, and prediction of coarse activity patterns [1,7,10]. Success is claimed only when the coarse graph matches or beats these baselines on tasks that a neuron-level model could also solve, and it fails when the collapse erases signal that only survives at single-neuron resolution.

- Sensorimotor routing: feed canonical sensory inputs and measure how faithfully coarse-type dynamics reproduce neuron-level output flow [10].
- Community recovery: verify that the graph re-discovers the isomorphic and sex-specific blocks found by hierarchical community detection [1].
- Activity prediction: fit coarse-type states to recorded population activity and compare explained variance against the neuron-level baseline [7].
- Drop synapse-count weighting and keep only edge presence to isolate the cost of aggregate magnitude.
- Remove neurotransmitter signs and compare unsigned versus signed coupling [1].
- Vary cell-type granularity from the supplied taxonomy up to SBM-defined blocks [1].
- Collapse the left and right hemispheres and measure the loss from removing the somaSide partitioning [1].

- Filter nodes by downstream capture to isolate the effect of poorly reconstructed partners [1].

What the coarse model can and cannot demonstrate

What it can demonstrate is coarse organizational structure: which cell types act as hubs, how signal routes between major neuropils, the magnitude of hemispheric asymmetry, and the relative contributions of isomorphic and dimorphic cell types [1]. It can also serve as a cheap, interpretable prior or null for the finer model classes, and as a check that single-neuron detail is not required to explain a given task. What it cannot demonstrate is anything that lives below the cell type: per-neuron tuning, within-type variability, precise spike timing, synaptic facilitation or depression, neuromodulatory state, and gap junctions, all of which the coarser graph deliberately averages out. It therefore establishes design constraints and aggregate motifs rather than mechanistic prediction, and any claim of neuron-level fidelity would be biologically unwarranted. The honest reading is that the coarse cell-type graph is a structural scaffold and a first cut at what a connectome can explain about information flow, not a model of how a *Drosophila* computation is actually realized.

Mesoscopic circuit model design

The mesoscopic tier sits between the coarse cell-type graph and a full neuron-level network: it keeps the major anatomical pathways of the MaleCNS intact while operating at the scale of neuron types rather than individual cells [11]. By construction it targets the circuit-level question of how the central nervous system turns sensory information and memory into motor action [1]. It therefore preserves the recurrence and topographic structure that a pooled cell-type graph discards, and it is the tier closest to existing connectome-constrained whole-brain implementations.

From connectivity matrix to reconstructed pathways

The source geometry is a single proofread male CNS reconstructed across seven enhanced focused ion beam scanning electron microscopy systems over thirteen months into a 160 teravoxel volume at $8 \times 8 \times 8 \text{ nm}$ isotropic resolution [1]. The reconstructed volume spans 0.082 mm^3 and is organized into colored CNS regions, with neuron counts reported per brain region, including sensory neurons [1]. It covers the whole central nervous system, so the model can route signal from optic-lobe input through central neuropils down into the ventral nerve cord and neck connective that carry motor output [1]. For each synapse the dataset reports per-region pre- and postsynaptic completion rates, and it reports the fraction of connections for which both partners are proofread, broken down by neuropil region [1].

Dynamics, cost, and training objective

$$\dot{x}(t) = -\lambda x(t) + W_h \sigma(W_{\text{rec}} x(t) + B u(t)) + I_{\text{ext}}$$

Baselines, evaluation, ablations, and success criteria

Design choices and validation plan for the mesoscopic circuit model

ASPECT	MESOSCOPIC DESIGN	JUSTIFICATION / SOURCE
Spatial scale	Neuron-type channels over the major CNS pathways	Mesoscopic level of neuron types [11]
Connectance	Curated major pathways retained from the proofread connectome	Proofreading targets connectivity coverage rather than completeness [1]
Synapse sign	Excitatory / inhibitory assigned from neurotransmitter classification	Neurotransmitter classification at synaptic sites [1]
Recurrence	Inherited recurrent topology, sensorimotor loop	CNS transforms sensory information into motor action [1]
Objective	Fit readout to measured neural activity	connectome-constrained networks fit and predict neural activity [1]
Baseline	Coarse cell-type graph, rewired and random connectomes	Structural controls alongside the prior coarse-graph tier
Evaluation	Reproduce sensorimotor circuit dynamics from the connectome	computational brain sensorimotor processing [1]
Ablation	Remove recurrence, permute synapse sign, or thin pathways	Isolate the contribution of each mesoscopic feature

Note

The mesoscopic tier trades biological detail for tractable recurrence: it reproduces sensorimotor circuit dynamics from the connectome, but it does not predict whole-animal behavior and it excludes the neuromodulatory, temporal, and learning-variable details reserved for the missing-variables discussion.

Neuron-level sparse recurrent model design

Building a neuron-level sparse recurrent model on the MaleCNS begins with the complete synaptic connectivity graph: the fully proofread adult male CNS was imaged with seven eFIB-SEM systems over thirteen months, producing a 160 teravoxel volume at $8 \times 8 \times 8$ nm isotropic resolution [1]. Each neuron's skeleton, synapse coordinates, and synapse counts are retrieved through neuPrint and the FlyWire synapse predictions at materialization version 783 [1]. In this graph the neurons are the recurrent units, so the weight matrix is a sparse per-neuron adjacency rather than a dense layer, and every synapse carries a spatial location that

becomes a fixed or learnable connection weight. The sign of each synapse is not directly visible in EM; instead an aggregate neurotransmitter prediction is assigned to each segment together with a per-segment confidence score derived from the formula in Eckstein et al [1]. Each presynapse within a segment then receives a confusion score drawn from a synapse-level confusion matrix, giving an explicit estimate of excitatory or inhibitory character plus uncertainty [1]. This reflects the broader result that neurotransmitter classification from EM images at synaptic sites is feasible, though synaptic identity cannot be read directly at full resolution [12].

Not every neuron is equally well characterized at the connectivity level: the downstream capture fraction — the share of a neuron's synaptic outputs that terminate on traced neurons — is bimodal across the central brain and shows where neuron-level connection recall degrades even when a cell is fully reconstructed [1]. For a neuron-level recurrent model this means the sparse connectivity must be reported with per-edge recall, because truncated downstream targets silently cut the recurrent signal.

These quality variations matter directly for the third proposed model class, because a neuron that is fully traced can still contribute a poorly characterized recurrent loop and so cannot be trusted as an unconstrained state variable [1].

Sparse recurrence, dynamics, and the inference problem

Because the per-neuron graph is sparse and densely recurrent, the natural embedding is a spiking or rate network with spatial recurrent dynamics driven by event-based inputs — the spiking neural network formulation that exploits location-based spiking to extract features from temporally structured data [13]. A connectome-constrained network of this kind satisfies three explicit constraints: the architecture is fixed by the connectome measurements, while the cellular and synaptic dynamics are supplied by a separate dynamical model, leaving the free weights to be trained against activity [7].

The decisive caution for success criteria comes directly from connectome-constrained recurrent networks: prediction of neural activity shows that a connectome often does not substantially constrain the dynamics of recurrent networks, illustrating the difficulty of inferring dynamics from structure alone [14]. A neuron-level model therefore cannot claim to reproduce CNS activity merely by wiring in the MaleCNS edges; the sparse recurrence fixes the permitted information flow, but the achieved dynamics are not guaranteed to match biology.

Note

Structure alone under-constrains dynamics: the same connectome-constrained recurrent networks show that a connectome often does not substantially constrain recurrent dynamics, so wiring in the MaleCNS edges does not by itself reproduce CNS activity [14].

Cost

The reconstruction itself was expensive: because the complete CNS was imaged and aligned as three stitched subvolumes of the right hemisphere, left hemisphere, and nerve cord, the segmentation procedure had to run three separate times [1]. At the model level, sparsity is the primary cost lever: a dense recurrence over all detected neurons scales with the full synapse count, so the trainable weight count must be bounded. One principled bound is to mask synapses below their transmitter-confidence score, using the same confidence signal that accompanies each segment's neurotransmitter prediction [1]. Running against the same segmentation and detector infrastructure used to build the connectome, including a fine-tuned T-bar detector and a lamina precision-recall pipeline, keeps data-preparation cost comparable to prior reconstructions [1].

Sensory and motor interfaces

The model must expose external sensory input and read out motor output, and maximum-flow analysis on the presynaptic-input-normalized graph identifies the possible information flow from source sensory neurons to target motor neurons subject to synaptic-capacity limits, with the neck connective identified as the constraining integration bottleneck [1]. Concrete anatomy-derived examples include DNp44 descending neurons carrying high maximum flow from olfactory and hygrosensory sources to many motor domains, and the identified first-, second-, and third-order thermosensory and hygrosensory neuron chains that carry modality signals toward the motor layer [1]. These provide a principled, anatomy-derived choice of which neurons are external inputs (the sensory subclasses) and which are observable outputs (the motor neurons), rather than an arbitrary interface split [1].

Training objective, baselines, and ablations

Baseline comparisons should include a connectome-free recurrent network of comparable size, the coarse cell-type graph model from the previous section, and a connectome-constrained architecture whose weights are frozen at the anatomical synapse counts.

1. Synaptic signs — comparing runs that use estimated transmitter signs with runs that use unsigned or sign-flipped synapses [1].
2. Connectivity completeness — using the downstream-capture metric to selectively mask low-recall edges and measuring the effect on dynamics reproduction [1].
3. Neuron inclusion — comparing full-neuron reconstruction with morphological co-clustering of similar neurons via NBLAST and low-dimensional UMAP embeddings to decide which cells can be collapsed without losing pathways [1].

Success criteria

Success would demonstrate that a sparse, anatomy-derived recurrent substrate can carry sensory-to-motor information and be trained toward activity while remaining faithful to the MaleCNS wiring, preserving the minimum flow from sensory sources to motor domains [1]. It would not demonstrate that the connectome alone determines CNS dynamics, given the established finding that structure under-constrains the recurrent behavior [14]. Concretely, a positive result shows that the interface wiring, transmitter-sign estimates, and sparse recurrence are sufficient for a task-relevant dynamical regime, while a negative or inconclusive result at least pins down which missing variables — transmitter confidence, downstream capture, or the free synaptic parameters — are responsible for any gap in activity reproduction [1].

State of connectome-inspired machine learning

From connectome to network architecture

Connectome-inspired machine learning proceeds from one empirical anchor: a biological connectome is a real connection pattern between neurons [15]. Work across the field treats that pattern in three ways — as the raw wiring of an artificial network, as a structural prior that regularizes training, and as a functional template for how large networks should be organized [15,16,17]. The successes cluster around small, fully sequenced organisms, while the failures cluster around the gap between static wiring and living computation.

Dominant approaches and their biological sources

The most direct line of work constructs artificial networks whose connectivity is taken wholesale from a biological connectome [15]. Building convolutional architecture on the basis of the nematode *Caenorhabditis elegans* connectome is one such attempt, motivated by the idea that biological wiring already encodes useful representational structure [15]. A second line reconstructs behavior from wiring by training a spiking neural network to reproduce *C. elegans* dynamics in the Elegans-AI framework, and more broadly constructs deep connectomics networks whose architecture is inspired by neuronal connectivity [16]. A third line shifts from anatomy to function, arguing that a functional connectome provides scalable ways to characterize the topology of large neural networks rather than relying on a static structural graph [17]. At the macroscale, machine learning applied to connectomes is framed around predictive models that carry out descriptive, predictive, and prescriptive analyses, using supervised algorithms to predict traits from connectome data [18].

1. Connectome-structured artificial networks build convolutional architecture directly from the nematode connectome [15].
2. Spiking reconstruction reproduces organism behavior with a *C. elegans*-weighted spiking neural network (Elegans-AI) and related deep connectomics networks [16].

3. Functional-connectome methods characterize the scalable topology of large networks from activity rather than anatomy [17].
4. Macroscale predictive models use supervised algorithms to predict traits through descriptive, predictive, and prescriptive analyses [18].

Recurring successes and recurring failures

The recurring success is that structural wiring can seed architecture or training, but its predictive and transfer value stays narrow. Most connectome-derived models succeed at descriptive or reconstruction tasks and weaken on genuine prediction and cross-system transfer. The recurring failure is the assumption that structural connectivity is itself functional. Deriving activity maps from electrical or electromagnetic sensors is contaminated by volume conduction, the effect whereby multiple sensors pick up activity from the same electrical source in the brain [19]. Because of that contamination, the wiring inferred from sensor data is not the synaptically resolved wiring that a faithful model requires [19]. Empirical connectomes therefore separate a structural scaffold from the dynamics that scaffold supports, and even well-validated connectome networks rarely close the gap between reconstruction and transfer.

Note

Volume conduction means multiple sensors record the same electrical source, so sensor-derived activity maps do not equal synaptically resolved connectivity [19].

Graph neural networks for neural connectivity

Graph neural networks (GNNs) form a distinct architecture family that operate directly on graph-structured data rather than on the regular grids of pixels or symbol sequences that dominate other ML. The canonical definition is a 'graph-in, graph-out' architecture, in which these models accept a graph as input with information loaded into its nodes and typically return a graph-structured output [20]. This maps almost literally onto a connectome: neurons become nodes, synapses and gap junctions become edges, and per-neuron state or per-synapse weight becomes the node and edge features. Because the algorithm treats the wiring itself as the input rather than requiring a fixed-size vector, the MaleCNS connectivity graph is exactly the kind of object a GNN is built to consume.

Message passing and the graph-in, graph-out paradigm

GNNs share a common computational backbone: iterative message passing, in which each node aggregates feature vectors from its neighbors and updates its own representation. In the network-neuroscience literature this step is described as a propagation rule that encapsulates both node and edge features in order to generate prediction scores [21]. The Distill survey frames the same mechanism as the defining property of the

architecture, noting that such models 'accept a graph as input, with information loaded into its nodes' [20]. For a connectivity-inspired model this means the number of cells and their pairing pattern can vary between instances without reshaping the network, which is the property a single architecture needs to span coarse cell-type, mesoscopic, and neuron-level connectome representations.

Encoding the brain as a graph

On the application side, graph networks have matured specifically as a tool for relational biological data. Graph Neural Networks (GNNs) have emerged as a novel paradigm that enables scientists to model complex relational data in medical [imaging] [22]. That principle transfers from population-level neuroimaging down to circuit scale, where the brain graph is taken to be a network whose nodes are regions or cells and whose edges encode structural or functional coupling [21]. Survey work in network neuroscience treats this propagation-based formulation as the standard route from raw connectivity to prediction, since the topology carries the relational information that dense or MLP encodings discard.

At the connectome scale the payoff is specificity rather than generality. While graph-based representations of neural networks are established, discovering the specific connectivity and activity patterns that predict burst initiation [23] remains an open task, which signals that the field is still moving from merely laying out a graph toward extracting the mechanisms encoded within it.

Design principles for connectivity-inspired networks

From these documented uses, several reusable design principles emerge for a connectome-inspired architecture.

- Edges can carry their own features, so synaptic sign and strength are not collapsed into the node representation; message passing explicitly encapsulates node and edge features [21].
- The same model family generalizes across unrelated structural domains rather than being fixed to one anatomy, since graph-network machinery has been reused outside neuroscience [24].
- The shared message-passing body can be pretrained and then fine-tuned on small biological datasets, which matters because connectomes are expensive to produce and verify [24].
- Dynamics must be layered on top of the static graph, because the research frontier is identifying which connectivity-and-activity patterns drive specific behaviors such as burst initiation rather than reconstructing the wiring alone [23].

Note

The core lesson for connectome-inspired design is that a GNN treats connectivity as the computation's domain rather than merely a regularizing prior: it preserves node and edge features through propagation [20,21] and has already proven portable to circuit-design tasks [24] while the hard scientific problem — finding which connectivity-and-activity patterns predict behavior — is exactly what these representations must still learn to expose [23].

Recurrent state-space and recurrent models

Any artificial model that preserves the recurrence of the insect central nervous system must pay for it, because a recurrent network's accuracy is tied directly to its sequential computational cost. Just as the neuron-level and mesoscopic tiers require recurrence to reproduce sensory-to-motor integration, the underlying ML question is how to achieve that recurrence cheaply. Two families answer it differently: structured state-space models (SSMs) that linearize the sequence dimension, and models that retain only a fraction of the biological feedback paths.

Structured state-space recurrences at long horizon

A structured state-space model replaces the hand-unrolled recurrent loop with a linear-time convolution over the input sequence. It was not until the Structured State Space Sequence Model (S4) that the efficiency and practical feasibility of structured SSMs were truly realized [25]. This is the turning point where the model's cost stops growing with the length of the simulated episode, because the recurrence collapses to a single convolution that can be computed in parallel across time [25]. The payoff is exactly the long-range retention that ordinary recurrent networks lose [26]. Traditional RNNs, by contrast, suffer well-known limitations including difficulty in effectively learning very long-range dependencies [26]. The accuracy cost of an SSM is that its core recurrence is effectively linear and context-independent, so it cannot dynamically route information based on the current input the way a gated recurrent unit can.

Continuous-time and liquid recurrence

Liquid neural networks are the continuous-time counterpart: each hidden unit obeys an ordinary differential equation whose few parameters define its dynamics. In a comparative study against traditional recurrent networks, liquid networks were positioned as a parameter-efficient and robust alternative precisely because the traditional RNNs struggle with long-range dependencies [26]. For a circuit-derived model, continuous-time recurrence is attractive on biological grounds, because a real recurrent circuit integrates synaptic input continuously and its state changes smoothly rather than in discrete jumps. The trade-off is a less mature training objective and a weaker formal guarantee on sequence length than the closed-form SSM scan.

Partial recurrence as a minimal recurrent circuit

Full biological recurrence is expensive and often unnecessary, so one can selectively keep only the feedback paths that the computation actually needs. Partially recurrent neural networks (pRNNs) retain a sparse set of recurrent connections and nonetheless compute robustly and efficiently on navigation-style maze tasks [27]. This is the practical sweet spot for an insect-analog model, where the recurrent path-integrator-like dynamics of the central complex can be approximated by a small fraction of the original feedback wiring. The cost-accuracy outcome is that pRNNs achieve much of a fully recurrent model's behavioral fidelity at a fraction of its sequential cost [27]. Their open question is how to choose which paths to keep without prior access to the biological computation each path supports.

Benchmarking against CNS dynamics and learning signals

The accuracy of any of these recurrent choices is best judged against real neural activity rather than against a synthetic task alone. Data-driven decoding approaches build powerful statistical models to assess how stimuli or behavior are encoded in recorded neural populations [28]. A recurrent model can be validated by fitting such a decoder to its own hidden states and measuring how faithfully the artificial recurrence reproduces the same stimulus or behavioral variables that the biological recording decodes [28]. There is also a biological regularization angle, because insect learning is delivered through dopaminergic neurons (DANs), which drive learning across the animal kingdom [29]. Because upstream circuits regulate the activity of those neurons, the natural substrate for such adaptive regulation is a recurrent architecture rather than a feed-forward one [29]. This suggests that the most faithful class couples the recurrent hidden dynamics with a modulatory teaching signal instead of relying on back-propagation-through-time alone.

Cost-accuracy tradeoff across recurrent families for insect-circuit mimicry

ARCHITECTURE	RECURRENCE FORM	SEQUENCE COST	LONG-RANGE HANDLING	INSECT-ANALOG FIT
Traditional RNN	Full recurrent hidden state	Sequential unroll; vanishing/exploding gradients [26]	Poor long-range retention [26]	Best for learning rules, costly to scale
Structured SSM (S4) [25]	Closed-form linear recurrence / selective convolution	Linear in sequence length, parallel scan [25]	Strong long-range retention [25]	Efficient, but linear core limits dynamic routing
Liquid neural network [26]	Continuous-time ODE hidden state	Low parameter count, moderate cost	Better than RNN on long-range [26]	Continuous-time matches biological integration
Partially RNN (pRNN) [27]	Sparse subset of feedback paths	Low sequential cost [27]	Robust on mazes [27]	Minimal recurrence for complex-like navigation [27]

Note

The core trade-off: SSMs win on long-horizon cost, liquid networks win on continuous-time biological fidelity, and pRNNs win on cost-per-behavior by keeping only the feedback that computation needs [25,26,27].

Circuit policies and biologically constrained RL

Two strategies let control agents inherit real circuits: neural-circuit policies (NCPs), which treat a connectome's layered wiring as a fixed structural prior, and biologically constrained reinforcement learning, which instead grounds the learning rule itself in mechanisms a brain could plausibly implement. The two are complementary: NCPs import the circuit's recurrence and signal flow into the architecture, while constrained RL constrains how that architecture is trained, and their combined goal is an agent whose dynamics are both interpretable and trainable on biologically relevant grounds.

Neural circuit policies as structural priors

NCPs employ a four-layer hierarchy that processes the information extracted through sensory neurons, forwarding it through an interneuron layer before driving motor neurons and closing the loop with proprioceptive feedback [30]. This layering mirrors an insect central nervous system in which sensory input is reshaped by central interneurons and only then converted into motor command [30]. Because the interneuron layer is recurrent, the policy's dynamics are not learned from scratch but inherit the time-dependent signal processing of the underlying circuit [31]. The payoff is an agent that learns coherent, transferable features

across diverse environments while remaining interpretable, since each layer maps onto a biological compartment and the fitted dynamics can be read back as sensory-to-motor processing [31]. In virtual character control this structure keeps the learned policy transparent as a sensory, central, and motor decomposition rather than an opaque end-to-end function [30].

Grounding the learning rule in biology

Architecture alone is not enough, because the policy must still be trained and it is during training that biological plausibility is most easily violated. The standard reinforcement-learning framework centers on a system in which an agent selects actions from an observable state and updates its policy via reward signals, components that map onto proposed neurobiological substrates such as dopaminergic value signals and their phasic form for credit assignment [32]. In practice, however, most artificial-agent work has concentrated on learning a single complex problem in a static environment, a setup that departs sharply from the dynamic, multi-task conditions a living organism must handle [33]. One route to closing this gap is to move to spiking neural networks, the third generation of artificial networks that more directly reproduce the temporal behavior of the biological brain, and to train them with local, brain-inspired learning rules suited to control tasks [34]. A spiking NCP trained under such local rules would preserve both the connectome-derived recurrence and the synaptic-plausibility constraint that global backpropagation through time fails to satisfy, which is the specific limit biologically constrained RL is designed to address [34].

Neural-circuit policies versus biologically constrained reinforcement learning

ASPECT	NEURAL CIRCUIT POLICIES	BIOLOGICALLY CONSTRAINED RL
Primary prior	Fixed sensory–interneuron–motor hierarchy [30]	Biologically plausible learning rule [34]
Signal flow	Recurrent interneuron layer with proprioceptive feedback [30]	Action selection from observed state and reward [32]
Interpretability	Each layer maps to a biological compartment [31]	Value, policy, and credit-assignment substrates [32]
Training scope	Coherent features across varied environments [31]	Largely single complex task in static environments [33]
Hardware class	Fully connected policy network [30]	Spiking neural network [34]

Missing variables and unwarranted claims

The MaleCNS reports which neurons connect to which and, at most, how many synapses span each pair of neurons [1]. It is therefore a wiring diagram rather than a functional specification, and every model built on it inherits the gap between the two. This section enumerates the variables the anatomy omits — synaptic physiology, neuromodulation, electrical coupling, temporal dynamics, developmental history, embodiment, and the learning rules that both built and maintain the connectome — and, for each proposed model class, assesses how each omission caps biological fidelity and ML utility. The guiding finding is that increasing anatomical faithfulness does not buy functional faithfulness: as coarse graphs become mesoscopic circuits and then neuron-level recurrences, models expose the missing variables rather than resolving them.

A static wiring diagram, not a dynamical specification

The published connection weights are defined as synapse counts, and these counts were distributed across the hemispheres using each source neuron's soma side as reference [1]. Unpaired medial neurons were split evenly between the two sides, and sensory neurons were routed by the side of the nerve through which their axons entered [1]. A synapse count is a structural proxy for a physiological quantity, not the quantity itself: it encodes no release probability, no short-term facilitation or depression, and no absolute or sign-correct conductance, so it cannot reproduce the postsynaptic integration a real synapse performs. One framing that exposes the gap is the max-flow analysis, which estimates the possible rate of information transfer from a sensory source to a motor target subject only to capacity limits set by these synapse-count weights [1]. That is a statement about topology and capacity, not about how a neuron responds to a spike train in time.

Neurotransmitter identity is likewise reported as a prediction rather than a measured, functional state: each segment was assigned an aggregate neurotransmitter prediction together with a confidence score, and each presynapse received a confusion score drawn from the associated synapse data [1]. Predicting a transmitter's category is not the same as knowing that it is being released, in what volume, or with what modulatory effect at a given moment. Neuromodulation's diffuse, state-dependent action is precisely the faculty that a static connectome cannot capture, which is why whole-brain, connectome-constrained modeling of neuromodulatory interventions such as deep brain stimulation must still supply the dynamics that the anatomy omits [35]. Gap junctions compound the problem: electron-microscopy volumes can resolve the membrane appositions corresponding to electrical coupling, but coupling conductance, rectification, and time-varying gating are not encoded in a synapse-count matrix, so any model built here inherits an implicit chemical-synapse-only assumption.

Developmental history, reconstruction error, and variable completeness

A wiring diagram also carries the developmental and technical history of the tissue imaged, and that history is uneven. The presence of a normally expected cell in the second optic lobe was attributed to a developmental error in one optic lobe, with a per-neuron error rate estimated below 1% [1]. Assuming independent errors across hemispheres, the chance of missing an entire cell type was estimated at less than 0.01%, roughly three pairs of neurons in the central brain, against more than 1,000 male-specific neurons observed [1]. Those headline numbers are reassuringly low, but they must be read against completeness at the connection level, where the relevant neuron-centric measure is 'downstream capture' — the fraction of a neuron's synaptic outputs that reach traced neurons [1]. A fully reconstructed neuron still has poorly characterized connectivity if its postsynaptic partners are incompletely reconstructed, and the central brain's downstream-capture distribution is itself bimodal, placing a large share of neurons in a lower-recall regime [1]. Reconstruction quality remained labor-bound even in the optic lobe, where lamina processing had to be retrained on new annotated cubes with a dedicated T-bar detector and where precision-recall reflects ongoing rather than finished work [1]. Quality control additionally relied on repairing false merges and tracing each soma to its cognate neuron, with untyped or unmatched neurons reinspected against hemispheric homologues [1]. The authors themselves limit comprehensive across-sex conclusions because the female nerve cord was only partially proofread, and note that the newer BANC dataset still awaits proofreading, joint cell typing, and a quantitative understanding of variation before sex differences can be drawn robustly [1]. The lesson is that 'complete' is a graded, compartment-specific property, and developmental-history effects — cells that should or should not exist, pathways that shifted during development — are not represented at all.

Temporal dynamics and learning rules

Connectomes are end-state snapshots, so they are silent on two mechanisms that most determine behavior: the timing of activity and the rules that revise the connectivity itself. Spiking dynamics can support temporal accumulation and adaptive computation, and reproducing them in engineered systems typically requires spike-based communication plus multi-factor learning rules rather than a simple error signal [36]. The MaleCNS provides the scaffold of which neuron connects to which, but not the membrane time constants, synaptic delays, or refractory structure that give that scaffold its temporal character, nor the plasticity rules — the same rules a developmental animal used to build the connectome and continues to use to adapt it. Grounding models in brain physiology explicitly highlights synaptic plasticity, sparse spike-based communication, and multimodal association as the functional layers that a bare connectome omits [37]. If the connectome is a structure that itself arose under learning and can be reshaped, then freezing it reproduces one moment along a plastic trajectory and cannot say how that trajectory was formed or how it would change [38].

Embodiment and the sensory-motor interface

Finally, a connectome is extracted from a body, so any model built on it implicitly assumes the sensory and motor apparatus the tissue once served. Information-flow analyses on the MaleCNS trace presynaptic-input-normalized flow from sensory neurons of a given modality to motor neurons of each motor-domain subclass — for instance, DNp44 descending neurons show high maximum flow to many motor domains from olfactory and hygro-sensory neurons [1]. That flow is defined entirely within the isolated nervous system, so the actual receptors, muscles, and peripheral feedback that the tissue once produced lie outside the data. A neuron-level or mesoscopic model can therefore reproduce a plausible sensory-to-motor mapping, but its inputs and outputs are boundary conditions imposed by the modeler rather than variables recovered from the connectome. The same caveat holds at every tier: the coarse graph, the mesoscopic circuit, and the neuron-level recurrence all inherit the assumption that a fixed sensory/motor interface can stand in for an embodied animal.

Fidelity gaps by model class

How missing variables interact with each proposed model class. Higher anatomical fidelity raises functional exposure rather than removing it.

MISSING VARIABLE	COARSE CELL-TYPE GRAPH	MESOSCOPIC CIRCUIT MODEL	NEURON-LEVEL SPARSE RECURRENCE
Synaptic physiology (weights = synapse counts) [1]	Cell-type averaged; physiology absent by construction [1]	Synapse-count weights on key pathways; no short-term dynamics [1]	Sign and weight estimates only; no conductance or dynamics [1]
Neuromodulation	Ignored at graph scale	Pathways labeled, but state-dependent modulation absent [35]	Static sign only; no modulatory state [35]
Gap junctions (electrical synapses)	Chemical-synapse-only assumption	Chemical-synapse-only assumption	Chemical-synapse-only assumption
Temporal dynamics	Coarse or absent recurrence	Mild recurrence; slow dynamics only	Requires assumed time constants and delays [36]
Developmental history and reconstruction error	Aggregation hides per-neuron errors	Error smoothed at the pathway level [1]	Per-neuron error and bimodal downstream-capture recall exposed [1]
Embodiment and peripheral feedback	Sensory/motor interfaces imposed by modeler [1]	Sensory/motor interfaces imposed by modeler [1]	Sensory/motor interfaces imposed by modeler [1]
Learning rules and plasticity	No update mechanism	No update mechanism	Fixed connectivity freezes plasticity [37]

Reading the table, as anatomical detail increases models become more exposed to the connectome's omissions rather than less. The coarse cell-type graph averages over physiology and error, so it is robust and cheap but faint and structureless in function; the neuron-level model is the richest, yet its fidelity is capped by exactly the variables the anatomy under-reports — the bimodal downstream-capture recall, the transient-versus-static distinction, and the plasticity rules it cannot encode [1,38]. An unwarranted claim would be to read any tier as demonstrating that a connectome alone determines behavior: what the MaleCNS constrains is topology, and everything from dynamics to embodiment must be added, assumed, or deliberately left out [37].

Note

Do not infer function from topology.

A connectome constrains which neuron can influence which, but the physiology, dynamics, modulators, and learning rules that turn that scaffold into a behavior generator are not in the data [1].

Silently assuming they are present — e.g., reading synapse counts as conductances, or treating the fixed connectome as if it were the trained result of development — is the specific unwarranted claim this survey is designed to prevent [37].

Synthesis

All three proposed model classes trace back to the same fully proofread MaleCNS reconstruction: a 160-teravoxel volume assembled across seven eFIB-SEM systems, organized as a sex-shared sensorimotor framework with a male-specific overlay concentrated in integrating centers [1]. The three tiers diverge not in their source material but in what each is willing to average out.

$$J_{ab} = \frac{W_{ab}^c}{n_a n_b}, \quad W_{ab}^c = \sum_{i \in a} \sum_{j \in b} W_{ij}$$

The coarse cell-type graph reduces N neurons to a small dense coupling matrix by summing synapse counts across cell types, buying simulation speed at the cost of every sub-cell-type detail [1]. The neuron-level sparse recurrent model instead keeps the per-neuron adjacency and the estimated excitatory-inhibitory synaptic signs [1]. The mesoscopic circuit sits between them, preserving the major pathways and recurrence while operating at neuron-type scale [11].

The defining tension of the report is that increasing anatomical faithfulness does not buy functional faithfulness. The coarse graph is cheap, robust, and honest about being a structural scaffold, yet it can demonstrate nothing that lives below the cell type and any neuron-level fidelity claim would be biologically unwarranted [1]. The neuron-level model is the richest, but it meets a more decisive constraint: connectome-constrained recurrent networks repeatedly show that structure does not substantially constrain dynamics, so wiring in the MaleCNS edges does not by itself reproduce central-nervous-system activity [14]. The mesoscopic

tier inherits the same ceiling, since it reproduces sensorimotor circuit dynamics yet does not predict whole-animal behavior [11]. A parallel tension runs through the state of connectome-inspired machine learning, where the field clusters around descriptive and reconstruction success but weakens on genuine prediction and cross-system transfer, precisely because structural connectivity is too often mistaken for function [15].

Three tiers trade faithfulness for tractability.

MODEL TIER	WHAT IT PRESERVES	WHAT IT CAPS
Coarse cell-type graph	A compact coupling matrix built from summed synapse counts across cell types [1].	All sub-cell-type detail, physiology, and dynamics.
Mesoscopic circuit	Major pathways and sensorimotor recurrence at neuron-type scale [11].	Whole-animal behavior, neuromodulation, and temporal learning variables.
Neuron-level sparse recurrence	Per-neuron sparsity and estimated excitatory-inhibitory synaptic signs [1].	Dynamics, because structure does not substantially constrain recurrence [14].

The available ML machinery mirrors this hierarchy. Graph neural networks consume the connectome as a graph-in, graph-out object that carries synaptic sign and strength through message passing and can span all three representation scales at once [21]. For recurrence, structured state-space models win on long-horizon cost, liquid neural networks win on continuous-time biological fidelity, and partially recurrent networks win on cost-per-behavior by keeping only the feedback paths the computation needs [25,26,27]. Control inherits the circuit both as a fixed structural prior — a sensory-interneuron-motor hierarchy with a recurrent interneuron layer — and as a biologically grounded learning rule, since dopamine-driven teaching signals favor local, brain-inspired rules over backpropagation-through-time, which falters in the static, single-task environments where most reinforcement learning currently sits [30,33,34].

Note

Read function from topology, not the other way around. The MaleCNS constrains which neuron can influence which, but synaptic physiology, neuromodulators, gap junctions, temporal constants, developmental history, and the learning rules that built the connectome are not in the data; reading synapse counts as conductances, or treating the fixed wiring as a trained developmental result, is the specific unwarranted claim this survey exists to prevent [1,37].

1. Treat each tier as a graded claim: the coarse graph earns its status only against a random connectome ensemble and a full neuron-level comparison, and its honest value is as an interpretable structural prior and null rather than mechanistic prediction [7,9].

2. Bound every fidelity claim to topology: because structure under-constrains dynamics and reconstruction completeness is graded and compartment-specific, success should read as 'a sparse, anatomy-derived substrate can carry sensory-to-motor information and be trained toward activity,' not as 'the connectome determines CNS behavior' [1,14].
3. Target the missing variables rather than assume them: the most faithful class couples connectome-derived recurrence with dopamine-driven teaching signals and local learning rules, and validates against recorded neural activity instead of synthetic tasks [28,29,34].

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